

# ECET 250E | Problem Set 5 · Capacitors and inductors

Solution key and walkthrough

Trace the connections first. Then choose the capacitor or inductor rule, substitute the values, and check the result.

Due October 13, 2026, 1:30 p.m. (checked in D2L October 7)

Ideal capacitors and uncoupled ideal inductors are assumed. Capacitors add in parallel; inductors add in series. A series junction must have no other branch. Keep the indicated voltage and current directions.

## 6.16 - Find the unknown capacitor

$$C = 20 \mu\text{F}.$$

## 6.17 - Three capacitor networks

(a) 3 F; (b) 8 F; (c) 1 F.

## 6.18 - Identical-capacitor network

$$C_{eq} = 24/11 \mu\text{F} \approx 2.18 \mu\text{F}.$$

## 6.26 - Charge and energy in parallel

$$C_{eq} = 35 \mu\text{F}; \text{ charges} = 750, 1500, 3000 \mu\text{C}; \text{ total energy} \approx 0.394 \text{ J}.$$

## 6.32 - Capacitor voltages from a current source

$$v_1(t) = 50 + 2083.333(1 - e^{-2t}) \text{ V}; v_2(t) = 20 + 416.667(1 - e^{-2t}) \text{ V}. \text{ At } 0.5 \text{ s: } W_{12} = 11.21 \text{ J}, W_{20} = 0.8031 \text{ J}, W_{40} = 1.606 \text{ J}.$$

## 6.40 - Voltage from a current graph

+25 V, 0 V, -25 V at 1, 3, 5 ms respectively.

## 6.43 - Energy stored in an inductor

$$W = 0.144 \text{ mJ} = 144 \mu\text{J}.$$

## 6.51 - Reduce an inductor network

$$L_{eq} \approx 7.78 \text{ mH}.$$

## 6.53 - Reduce a network with diagonal branches

$$L_{eq} = 20 \text{ mH}.$$

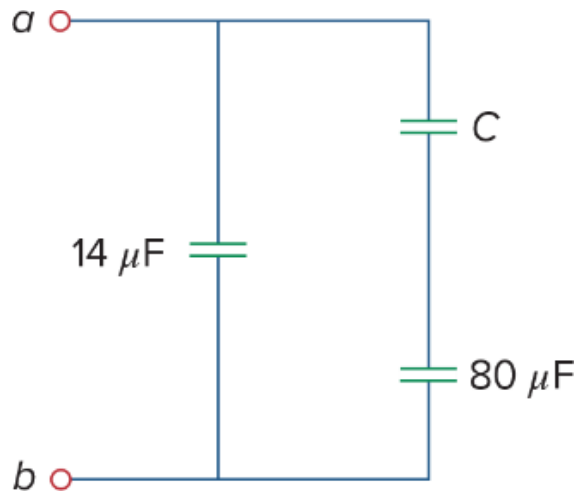
## 6.60 - Current sharing between parallel inductors

$$i_o(t) = 0.5 + 1.5e^{-2t} \text{ A}; v_o(t) = -15e^{-2t} \text{ V}.$$

## 6.16 - Find the unknown capacitor

Given / find: The equivalent at a-b is  $30\ \mu\text{F}$ . A  $14\ \mu\text{F}$  branch is in parallel with C in series with  $80\ \mu\text{F}$ .

Why this method: Separate the parallel branches before solving the unknown series pair.



### 1. Remove the known parallel branch

Parallel capacitances add. The C- $80\ \mu\text{F}$  branch must supply the difference.

$$C_{\text{branch}} = 30 - 14 = 16\ \mu\text{F}$$

### 2. Write the series relation

Use reciprocal addition, or the product-over-sum form for two capacitors. Keep all capacitances in  $\mu\text{F}$ .

$$16 = \frac{80C}{80 + C}$$

### 3. Solve for C

Multiply by the denominator, collect the C terms, then divide.

$$1280 + 16C = 80C \Rightarrow C = 20\ \mu\text{F}$$

Answer:  $C = 20\ \mu\text{F}$ .

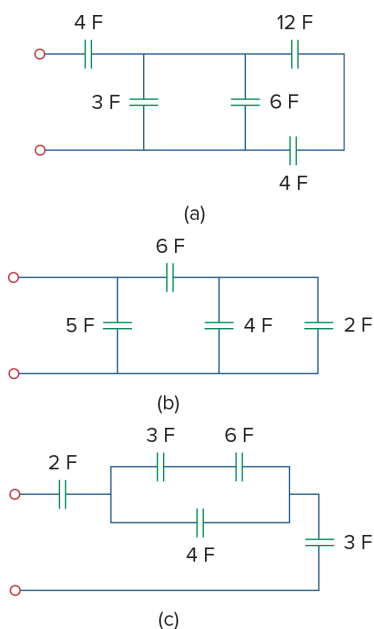
Check:  $20 \times 80 / (20 + 80) = 16\ \mu\text{F}$ ; adding the  $14\ \mu\text{F}$  branch gives  $30\ \mu\text{F}$ .

Common mistake: Adding C and 80 directly. They are in series, so their equivalent is smaller than either one.

## 6.17 - Three capacitor networks

Given / find: Find the equivalent capacitance in Fig. 6.51(a), (b), and (c). All labels in these three diagrams are in F.

Why this method: Reduce the innermost simple pairs, then redraw the remaining connections.



### 1. (a) Start at the right-hand branch

The 12 F and lower 4 F meet at an isolated right-hand node, so they are in series.

$$C_{right} = \frac{12(4)}{12+4} = 3 \text{ F}$$

### 2. (a) Combine across the middle nodes

That 3 F equivalent is parallel with the central 3 F and 6 F. The result is in series with the input 4 F.

$$C_a = \frac{4(3+3+6)}{4+(3+3+6)} = 3 \text{ F}$$

### 3. (b) Work from right to left

The right-hand 4 F and 2 F are parallel: 6 F. This is in series with the top 6 F, giving 3 F. The left 5 F is parallel with that whole branch.

$$C_b = 5 + \frac{6(4+2)}{6+(4+2)} = 8 \text{ F}$$

### 4. (c) Reduce the rectangle

The top 3 F and 6 F are series: 2 F. Add the lower 4 F in parallel to get 6 F. The remaining 2 F, 6 F and right-hand 3 F are in series.

$$\frac{1}{C_c} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1 \Rightarrow C_c = 1 \text{ F}$$

Answer: (a) 3 F; (b) 8 F; (c) 1 F.

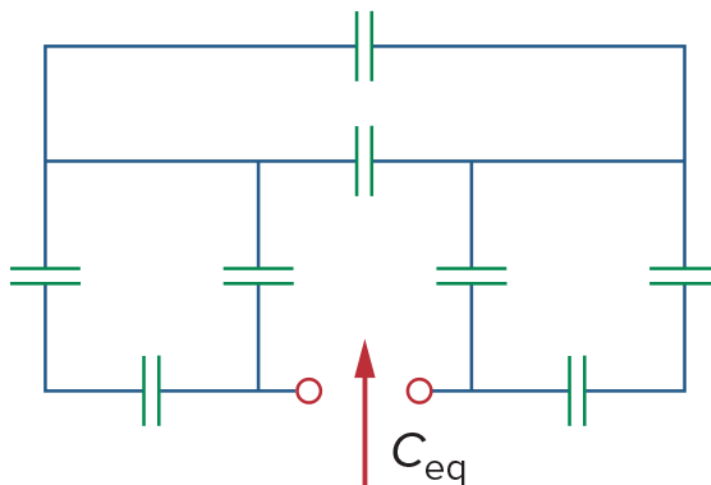
Check: The series result in (a) is below 4 F; (b) exceeds its direct 5 F branch; (c) is below its smallest series member, 2 F.

Common mistake: Assuming nearby components are parallel. They must share both endpoints.

## 6.18 - Identical-capacitor network

Given / find: Every capacitor in Fig. 6.52 is  $4\ \mu\text{F}$ . Find the equivalent between the two bottom terminals.

Why this method: Name the upper-left and upper-right nodes. The outer branches on each side reduce separately, leaving three equivalents in series.



### 1. Reduce the left side

The outer vertical  $4\ \mu\text{F}$  and bottom-left  $4\ \mu\text{F}$  form a series path from the left terminal to the upper-left node. Their  $2\ \mu\text{F}$  equivalent is parallel with the inner vertical  $4\ \mu\text{F}$ .

$$C_L = \frac{4(4)}{4+4} + 4 = 6\ \mu\text{F}$$

### 2. Reduce the right side

The right side has the same connections, so its equivalent is also  $6\ \mu\text{F}$ .

$$C_R = 6\ \mu\text{F}$$

### 3. Reduce the top connection

The topmost and middle horizontal capacitors connect the same upper-left and upper-right nodes. They are parallel.

$$C_T = 4 + 4 = 8\ \mu\text{F}$$

### 4. Combine the remaining chain

The reduced path between the bottom terminals is  $6\ \mu\text{F}$ , then  $8\ \mu\text{F}$ , then  $6\ \mu\text{F}$  in series.

$$C_{eq} = \left( \frac{1}{6} + \frac{1}{8} + \frac{1}{6} \right)^{-1} = \frac{24}{11}\ \mu\text{F}$$

Answer:  $C_{eq} = 24/11\ \mu\text{F} \approx 2.18\ \mu\text{F}$ .

Check: The result is smaller than each of the three series equivalents. Trace the path between the two open bottom terminals.

Common mistake: Treating the two bottom terminals as connected by a wire; they are the measurement port, not a short circuit.

## 6.26 - Charge and energy in parallel

Given / find: 5  $\mu\text{F}$ , 10  $\mu\text{F}$ , and 20  $\mu\text{F}$  capacitors are connected in parallel across 150 V. Find total capacitance, each charge, and total energy.

Why this method: Parallel branches all have the same 150 V. Use  $q = CV$  for each branch and  $W = \frac{1}{2}CV^2$  for energy.

### 1. Add capacitances

For parallel capacitors, add their values.

$$C_{eq} = 5 + 10 + 20 = 35 \mu\text{F}$$

### 2. Calculate each charge

Multiply each capacitance by the common voltage. Using  $\mu\text{F} \times \text{V}$  gives  $\mu\text{C}$ .

$$q_1 = 750 \mu\text{C}, \quad q_2 = 1500 \mu\text{C}, \quad q_3 = 3000 \mu\text{C}$$

### 3. Calculate total stored energy

Use the total capacitance in farads, and square the voltage.

$$W = \frac{1}{2}(35 \times 10^{-6})(150)^2 = 0.39375 \text{ J}$$

Answer:  $C_{eq} = 35 \mu\text{F}$ ; charges = 750, 1500, 3000  $\mu\text{C}$ ; total energy  $\approx 0.394 \text{ J}$ .

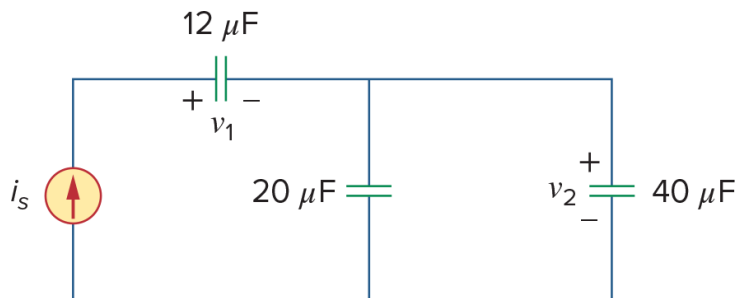
Check: Total charge = 5250  $\mu\text{C} = C_{eq} \times 150 \text{ V}$ . Individual energies 0.05625, 0.1125, and 0.225 J add to 0.39375 J.

Common mistake: Dividing the 150 V between parallel branches. Each receives the full source voltage.

## 6.32 - Capacitor voltages from a current source

Given / find:  $i_s = 50e^{-2t}$  mA,  $v_1(0) = 50$  V,  $v_2(0) = 20$  V. The source drives  $12\ \mu\text{F}$  in series with parallel  $20\ \mu\text{F}$  and  $40\ \mu\text{F}$ . Find  $v_1(t)$ ,  $v_2(t)$ , and energy in every capacitor at  $0.5$  s.

Why this method: Current entering a capacitor's positive terminal increases its voltage. Integrate current to get added charge; divide that charge by the appropriate capacitance.



### 1. Identify the shared current and voltage

The  $12\ \mu\text{F}$  carries all source current. The  $20\ \mu\text{F}$  and  $40\ \mu\text{F}$  are parallel, so both have  $v_2$  and behave as  $60\ \mu\text{F}$  for finding  $v_2$ . The indicated current enters the positive side of  $v_1$ .

### 2. Integrate the source current from 0 to t

Convert  $50$  mA to  $0.050$  A. The integral is charge in coulombs.

$$\Delta q = \int_0^t 0.050 e^{-2\tau} d\tau = 0.025(1 - e^{-2t})\ \text{C}$$

### 3. Add the initial voltages

Apply  $v(t) = v(0) + \Delta q/C$  separately to the  $12\ \mu\text{F}$  capacitor and the  $60\ \mu\text{F}$  parallel equivalent.

$$v_1 = 50 + 2083.333(1 - e^{-2t})\ \text{V}$$

### 4. Write the shared parallel voltage

This is the voltage of both the  $20\ \mu\text{F}$  and the  $40\ \mu\text{F}$  capacitor.

$$v_2 = 20 + 416.667(1 - e^{-2t})\ \text{V}$$

### 5. Evaluate at half a second

Use  $e^{-1} = 0.367879$ . The voltages are  $v_1 = 1366.918$  V and  $v_2 = 283.384$  V.

### 6. Find energy in each physical capacitor

Use its own capacitance. The  $60\ \mu\text{F}$  equivalent is useful for voltage but the question asks for energy in each capacitor.

$$W_C = \frac{1}{2} C v^2$$

Answer:  $v_1(t) = 50 + 2083.333(1 - e^{-2t})$  V;  $v_2(t) = 20 + 416.667(1 - e^{-2t})$  V. At  $0.5$  s:  $W_{12} = 11.21$  J,  $W_{20} = 0.8031$  J,  $W_{40} = 1.606$  J.

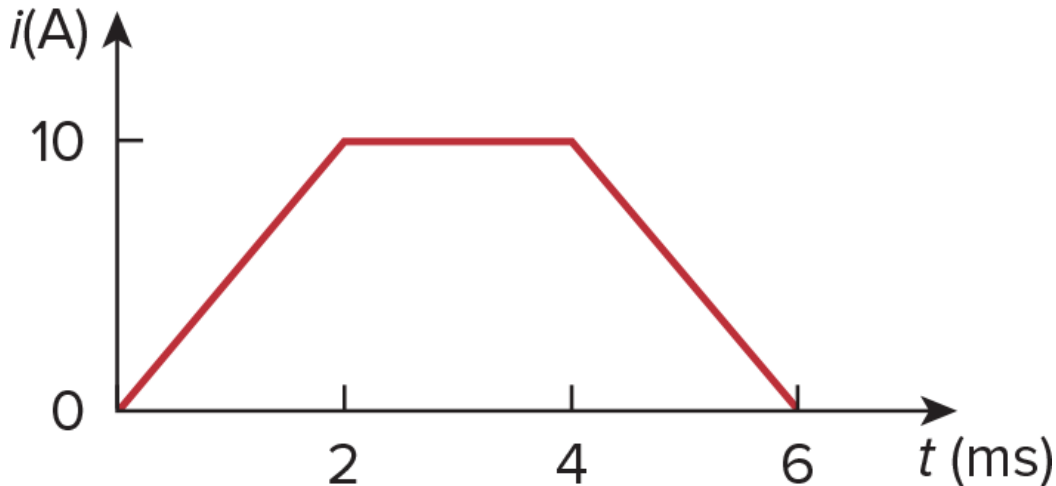
Check: At  $t = 0$  both formulas recover the given voltages. The  $40\ \mu\text{F}$  stores twice the energy of the  $20\ \mu\text{F}$  because they share the same voltage. The large voltages follow from the ideal mA source charging  $\mu\text{F}$  capacitors.

Common mistake: Dropping the initial voltage, using  $50$  A instead of  $0.050$  A, or dividing the full source current by each parallel capacitance separately.

## 6.40 - Voltage from a current graph

Given / find: A 5 mH inductor has a current ramp 0 to 10 A over 0–2 ms, constant 10 A over 2–4 ms, and a ramp back to zero over 4–6 ms. Find voltage at 1, 3, and 5 ms.

Why this method: Inductor voltage depends on slope, not current magnitude. Use the passive convention: current enters the positive-voltage terminal.



### 1. Use the slope of each straight segment

The relation is  $v = L \, di/dt$ . Convert  $L$  to 0.005 H and each 2 ms interval to 0.002 s.

$$v = L \frac{di}{dt}$$

### 2. At 1 ms: rising ramp

The slope is  $(10 - 0)/0.002 = 5000 \text{ A/s}$ .

$$v(1 \text{ ms}) = (0.005)(5000) = 25 \text{ V}$$

### 3. At 3 ms: flat segment

The current is nonzero but unchanging, so  $di/dt$  is zero.

$$v(3 \text{ ms}) = 0 \text{ V}$$

### 4. At 5 ms: falling ramp

The slope is  $(0 - 10)/0.002 = -5000 \text{ A/s}$ .

$$v(5 \text{ ms}) = -25 \text{ V}$$

Answer: +25 V, 0 V, -25 V at 1, 3, 5 ms respectively.

Check: Positive voltage increases current; negative voltage decreases it. The requested times are away from the graph's corners.

Common mistake: Using 5 A/ms directly with henries. Convert to 5000 A/s first.

## 6.43 - Energy stored in an inductor

Given / find: An 80 mH inductor's current rises from zero to 60 mA. Find its final stored energy.

Why this method: Inductor energy depends on the final current squared, regardless of how quickly it got there.

### 1. Convert the units

80 mH = 0.080 H; 60 mA = 0.060 A.

### 2. Square the current and multiply

Use  $W = \frac{1}{2}LI^2$ .

$$W = \frac{1}{2}(0.080)(0.060)^2 = 1.44 \times 10^{-4} \text{ J}$$

Answer:  $W = 0.144 \text{ mJ} = 144 \text{ } \mu\text{J}$ .

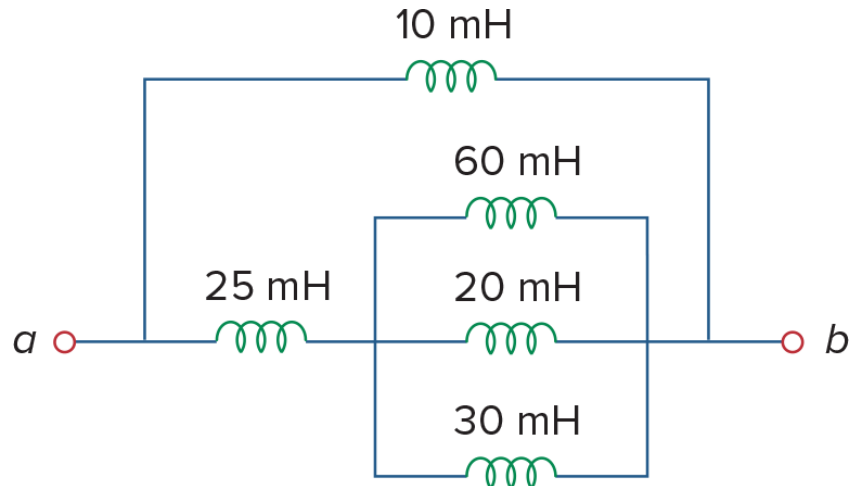
Check: The energy is nonnegative. At steady current an ideal inductor has zero voltage but can still store energy.

Common mistake: Concluding that steady state means zero energy, or forgetting that the current is squared.

## 6.51 - Reduce an inductor network

Given / find: Find  $L_{eq}$  at a-b in Fig. 6.73: a 10 mH top branch in parallel with 25 mH in series with a parallel 60, 20, 30 mH group.

Why this method: For uncoupled inductors, the series/parallel rules match resistors: series values add; parallel reciprocals add.



### 1. Reduce the right-hand parallel group

The 60, 20, and 30 mH inductors share both endpoints.

$$L_P = \left( \frac{1}{60} + \frac{1}{20} + \frac{1}{30} \right)^{-1} = 10 \text{ mH}$$

### 2. Add the series 25 mH

The lower path now has 25 mH + 10 mH.

$$L_{lower} = 35 \text{ mH}$$

### 3. Combine the two a-b branches

The top 10 mH is parallel with the entire 35 mH lower path.

$$L_{eq} = \frac{10(35)}{10 + 35} = \frac{70}{9} \text{ mH}$$

Answer:  $L_{eq} \approx 7.78 \text{ mH}$ .

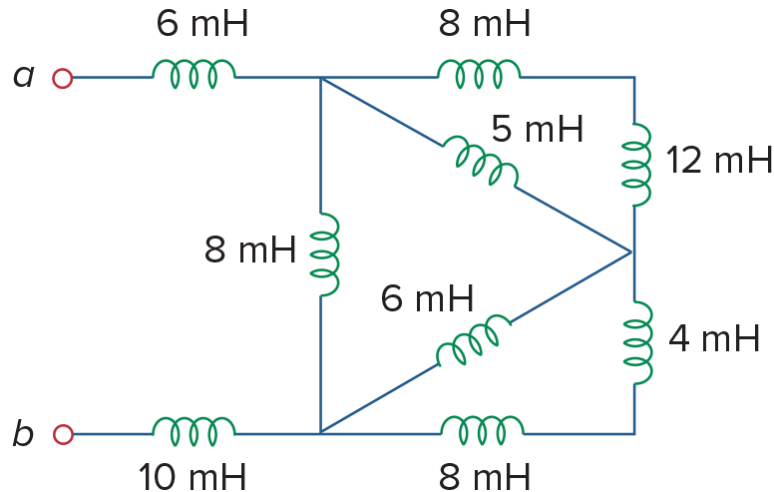
Check: A parallel equivalent must be smaller than its smallest branch, here 10 mH.

Common mistake: Putting the top 10 mH in series with 25 mH; a branch junction lies between them.

## 6.53 - Reduce a network with diagonal branches

Given / find: Find  $L_{eq}$  for Fig. 6.75. Name the internal top-left node U, bottom-left node D, and right-middle node M.

Why this method: Diagonal drawing lines do not require a special transform here. The right-hand outside pairs each have an isolated intermediate node.



### 1. Reduce U to M

The upper 8 mH and right-side 12 mH are series: 20 mH. This path is parallel with the 5 mH diagonal.

$$L_{UM} = 20 \parallel 5 = \frac{20(5)}{25} = 4 \text{ mH}$$

### 2. Reduce D to M

The lower 8 mH and right-side 4 mH are series: 12 mH. This is parallel with the 6 mH diagonal.

$$L_{DM} = 12 \parallel 6 = 4 \text{ mH}$$

### 3. Reduce U to D

The route through M has  $4 + 4 = 8$  mH. It is parallel with the left internal 8 mH.

$$L_{UD} = 8 \parallel 8 = 4 \text{ mH}$$

### 4. Add the terminal inductors

The 6 mH at terminal a and 10 mH at terminal b are in series with the reduced internal network.

$$L_{eq} = 6 + 4 + 10 = 20 \text{ mH}$$

Answer:  $L_{eq} = 20 \text{ mH}$ .

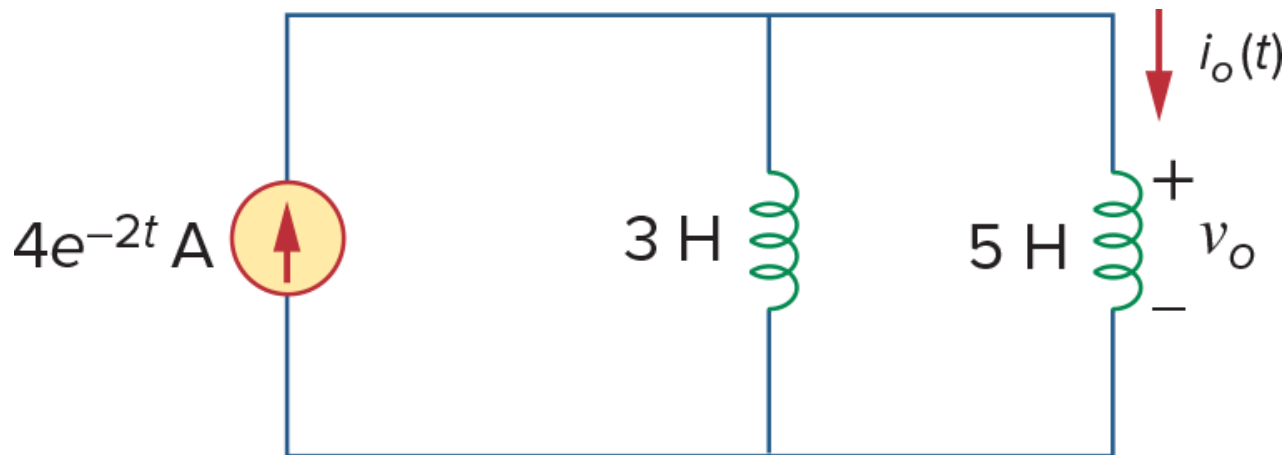
Check: The two terminal inductors alone contribute 16 mH. The internal network adds 4 mH. Here  $\parallel$  denotes parallel combination.

Common mistake: Treating the two diagonal inductors as series before reducing the outer paths: their shared node initially has additional branches.

## 6.60 - Current sharing between parallel inductors

Given / find: A source  $i_s(t) = 4e^{-2t}$  A drives parallel 3 H and 5 H inductors.  $i_o$  points downward through 5 H, with  $i_o(0) = 2$  A.  $v_o$  is positive at the top. Find  $i_o(t)$  and  $v_o(t)$  for  $t > 0$ .

Why this method: The two inductors share voltage. Use KCL on the derivatives to find that voltage, then integrate the 5 H branch current using its initial value.



### 1. Differentiate Kirchhoff's current law

Take both branch currents downward.  $i_s = i_3 + i_o$ , and  $di_3/dt = v_o/3$ ,  $di_o/dt = v_o/5$ .

$$\frac{di_s}{dt} = v_o \left( \frac{1}{3} + \frac{1}{5} \right)$$

### 2. Find the common voltage

The equivalent is  $(1/3 + 1/5)^{-1} = 15/8$  H. Differentiate  $4e^{-2t}$  to get  $-8e^{-2t}$  A/s.

$$v_o = \frac{15}{8}(-8e^{-2t}) = -15e^{-2t} \text{ V}$$

### 3. Integrate the 5 H branch slope

Divide the common voltage by 5 H, then integrate from 0 to  $t$  and add  $i_o(0) = 2$  A.

$$i_o = 2 + \int_0^t -3e^{-2\tau} d\tau = 0.5 + 1.5e^{-2t} \text{ A}$$

### 4. Check the other branch

KCL gives  $i_3 = 4e^{-2t} - i_o = 2.5e^{-2t} - 0.5$  A. At long times the branches carry opposite 0.5 A currents even as the source current tends to zero.

Answer:  $i_o(t) = 0.5 + 1.5e^{-2t}$  A;  $v_o(t) = -15e^{-2t}$  V.

Check:  $i_o(0) = 2$  A. Also  $5 di_o/dt = -15e^{-2t}$  V. The remaining circulating current is allowed by the ideal lossless-inductor model.

Common mistake: Using an instantaneous current-divider formula and losing the initial conditions. That shortcut omits the constant 0.5 A term.

Source: course handout linked from the online walkthrough. Independent study key, not an instructor-issued answer key.